Model Card and Explainability Architecture

Intended use, uses ruled out, data, performance, calibration, stability, how a decision is explained and audited, findings and limitations.

Version 1.0 October 2025 Prince Akonor Asare

1. Model identification

Model name	Alternative Credit Assessment Scorecard, thin file variant
Model identifier	GCR-SC-001
Version	1.0
Model type	Logistic regression on Weight of Evidence transformed characteristics
Risk tier	Tier 1, decision influencing
Owner	Get Credit Rate Foundation
Status	Phase 1 development complete. Not approved for live decisions.

2. Intended use

The model produces an ordinal risk score and a set of principal reasons for adverse action, for use by a certified Community Development Financial Institution in assessing applicants who lack sufficient credit bureau history to be scored by conventional models.

The model is intended to run alongside an existing underwriting process and to inform a human credit decision. It is not intended to render an automated decline without human review.

3. Out of scope uses

The model must not be used for any of the following.

1. Automated decline without human review.
2. Pricing or rate setting. The model was not developed against loss given default or profitability outcomes.
3. Decisions at any institution before coefficients are refit on that institution portfolio.
4. Any protected class inference. No protected class variable is admitted to the feature set.
5. Collections, account management or line assignment.

4. Training and evaluation data

Source	Public consumer loan performance file, accepted originations 2007 through 2018 quarter four
Source rows	2,260,701
Loans used	1,249,246, vintages 2012 to 2017
Realized defaults	254,055 in the loans used, 268,599 in all matured loans



Target definition	Loan resolved as charged off or defaulted, against resolved as paid
Modelling window	Origination vintages 2012 through 2017
Training set	786,820 loans, vintages 2012 through 2015, default rate 18.64 percent
Test set	462,426 loans, vintages 2016 through 2017, default rate 23.23 percent
Split design	Out of time. Later vintages held out entirely.

Excluded from every feature set to prevent target leakage: all post origination payment, recovery, settlement and hardship fields, the refreshed bureau score, and the originating platform internal risk grade, sub grade and interest rate. The internal risk grade was retained solely as an independent benchmark on the test set.

5. Characteristics

31 characteristics after removal of the bureau credit score and all long horizon tradeline history. Weight of Evidence bins were fitted on the training vintages only and applied unchanged to the test vintages. Bin tables and information values are reproduced in Appendix A of the validation report.

6. Performance

Measure	Value
Out of time AUC, score and history removed	0.6883
Out of time Gini	0.3766
Kolmogorov Smirnov statistic	0.2707
Out of time AUC, full file variant	0.6957
Out of time AUC, incumbent internal grade	0.6874

Removing the bureau credit score and all long horizon repayment history reduced ranking accuracy by 0.0074 AUC. The reduced feature set matched the incumbent grade at plus 0.0009 AUC on the same withheld loans. That match is not evidence of capability on a credit invisible applicant, because the reduced set retains current credit report fields. Removing every credit report field leaves eight application level characteristics at 0.6535 AUC, which is 81.9 percent of the incumbent Gini.

Score band	Loans	Realized default rate
540 and below	68,782	44.61%
540 to 560	128,682	29.13%
560 to 580	162,276	18.40%
580 to 600	82,206	10.26%
600 to 620	18,955	4.96%
620 to 640	1,512	2.12%

Scaling uses 20 points to double the odds, a base score of 600 at 20 to 1 odds, factor 28.8539



and offset 513.5614. Model intercept is negative 1.476631.

7. Calibration

Predicted probability understates realized default in every decile of the held out set. The lowest risk decile predicted 4.78 percent against 6.48 percent realized. The highest risk decile predicted 42.36 percent against 48.09 percent realized. Deciles are strictly monotonic in realized default rate with no reversals.

The cause is identifiable. The training era default rate was 18.64 percent and the test era rate was 23.23 percent. Rank ordering transfers out of time. The intercept requires recalibration on any deployment population before the score is used in a live decision.

8. Stability

Population stability index between the training score distribution and the held out score distribution is 0.0081, below the 0.10 threshold conventionally treated as stable. Per vintage values range from 0.0008 to 0.0322.

The combination of a stable score distribution and drifted calibration indicates that the applicant population did not shift while the default environment worsened. This distinguishes recalibration of the intercept from refitting the model.

9. Explainability architecture

The model is a linear scorecard on binned characteristics. Every score is decomposable into a point contribution per characteristic, and every point contribution traces to a documented bin with a published Weight of Evidence value. No component of the score is derived from a method that cannot be inspected.

Adverse action reasons are produced by ranking, for each applicant, the points lost on each characteristic relative to the highest scoring bin for that characteristic. The four largest deficits are returned as the principal reasons. Reason text is drawn from a fixed mapping table



and is not generated per decision.

The audit record retained for each decision comprises the input feature vector, the bin assigned to each characteristic, the points awarded per characteristic, the total score, the model version identifier, the bin table version identifier, and the timestamp.

10. Findings from Phase 1 testing

Three findings were produced by exercising the model and reason code engine on the held out vintages and are

recorded here as open items.

Finding 1. Requested loan term appears as a principal reason in 99.8 percent of the lowest scoring 5,000 records. Term carries the highest information value at 0.2448 and a 29.8 point range. Term is a product selection, and an adverse action reason citing the term the applicant applied for is circular. Resolution is to fit separate scorecards by term, which is standard practice, or to remove term and treat it as a policy and pricing dimension.

Finding 2. A missing employment length value surfaced as a principal reason in 13.1 percent of the lowest scoring records. A missing data indicator is not acceptable adverse action notice language. Resolution is a revised reason mapping that either states the underlying condition in plain language or excludes the characteristic from reason generation.

Finding 3. One characteristic, total installment high credit limit, carries a coefficient sign opposite to its Weight of Evidence direction. This indicates multicollinearity. Resolution is to apply sign constraints and drop characteristics that retain a counterintuitive coefficient.

11. Limitations

1. No cash flow, rental, utility or telecom variable exists in the source. Alternative data feature validation is performed at the Phase 3 calibration stage on each institution’s records.
2. The source contains no race, ethnicity or sex field, so disparate impact could not be tested at this phase.
3. Vintages from 2018 were excluded because term length exceeds the observation window.
4. Test vintages are not fully matured for 60 month terms.
5. The applicant population is not a CDFI applicant population.
6. This population is credit visible. Variants B and D reduce the feature set to approximate an applicant with limited or no credit file. They do not replicate one. Variant B in particular retains current credit report fields, so its parity with the benchmark is not evidence of capability on a credit invisible applicant.

12. Monitoring and review



Population stability index	Quarterly, against the development distribution
Calibration	Quarterly, predicted against realized by score band
Ranking accuracy	Annually, on a rolling out of time window
Reason code distribution	Quarterly, for concentration and notice quality
Full revalidation	Annually, or on any material change to inputs or population

13. Ownership and approval

Developed by	Get Credit Rate Foundation
Independent review	Commissioned at each Phase 3 calibration under the Phase 2 scope of work. Compensating control recorded in the model risk management policy.
Approved for live use	No. Approval requires refitting on a partner institution portfolio, disparate impact testing, and resolution of the findings in section 10.